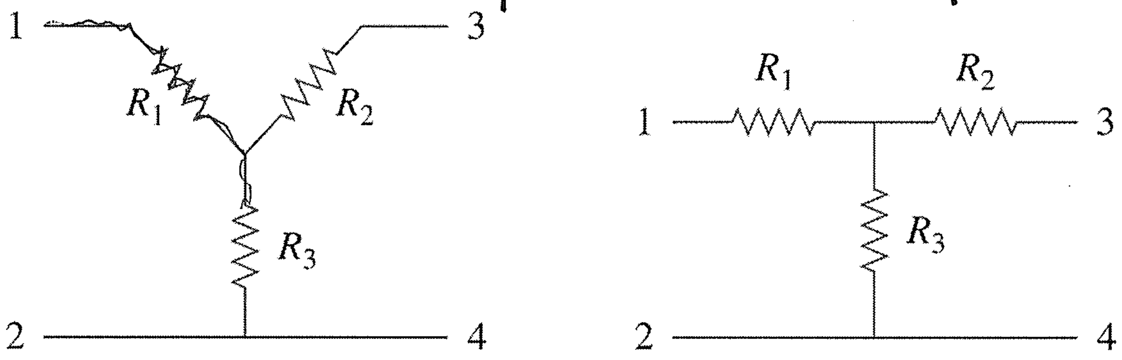
L#06: Wye-Delta transformations

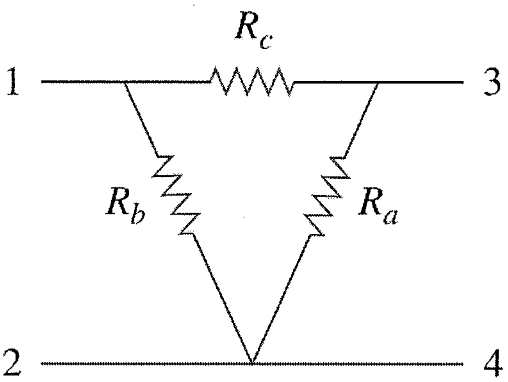
2018-02-15

Y $\longleftrightarrow$ T



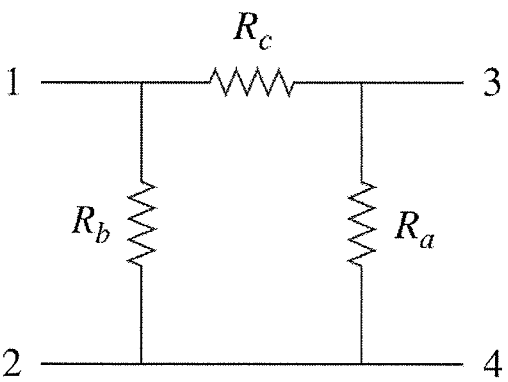
(a) (b)

Δ



(a)

π



(b)

$$R_{bc} = R_a \,||\, (R_b + R_c) = R_2 + R_3$$

$$\Longrightarrow R_2 + R_3 = \frac{R_a(R_b + R_c)}{R_a + R_b + R_c} \quad (1)$$

$$R_{ab} = R_c \,||\, (R_a + R_b) = R_1 + R_2$$

$$R_1 + R_2 = \frac{R_c(R_a + R_b)}{R_a + R_b + R_c} \quad (2)$$

$$R_1 + R_3 = \frac{R_b(R_a + R_c)}{R_a + R_b + R_c} \quad (3)$$

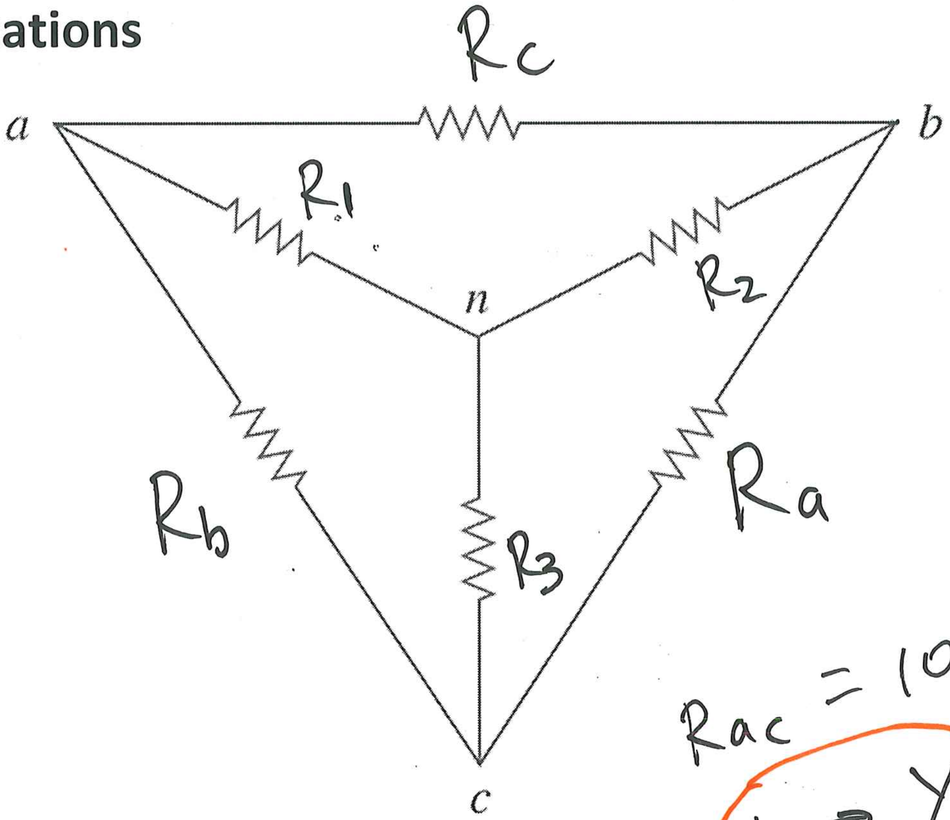
(2) + (3) − (1)

$$2R_1 = \frac{\cancel{R_a R_c} + R_b R_c + \cancel{R_a R_b} + R_b R_c - \cancel{R_a R_b} - \cancel{R_a R_c}}{R_a + R_b + R_c}$$

$$\cancel{2}R_1 = \frac{\cancel{2} R_b R_c}{R_a + R_b + R_c}$$

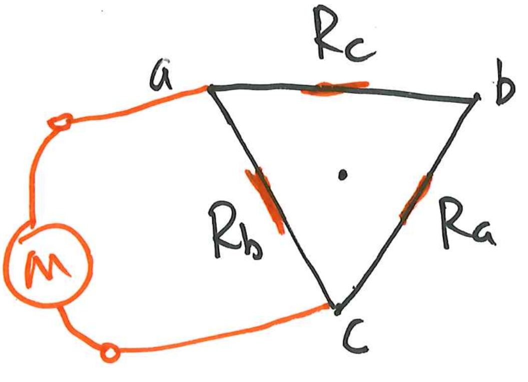
$$R_1 = \frac{R_b R_c}{R_a + R_b + R_c}$$

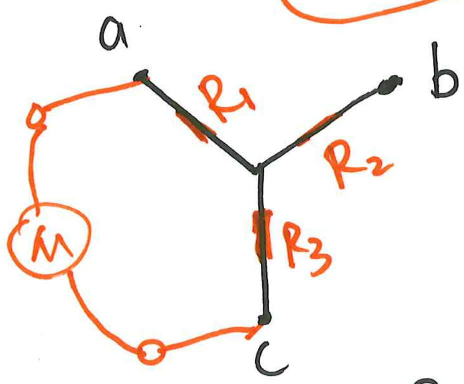
Notations



$R_{ac} = 100\,\Omega$

$\Delta \rightarrow Y$





$R_{ac} = R_b \,||\, (R_a + R_c)$, $R_{ac} = R_1 + R_3$

$$R_1 + R_3 = \frac{R_b(R_a + R_c)}{R_a + R_b + R_c}$$

$$R_1 = \frac{R_b R_c}{R_a+R_b+R_c} \; ; \qquad R_2 = \frac{R_a R_c}{R_a+R_b+R_c}$$

$$R_3 = \frac{R_a R_b}{R_a+R_b+R_c}$$

$$R_1 R_2 = \frac{R_a R_b R_c^2}{(R_a+R_b+R_c)^2}$$

$$R_1 R_3 = \frac{R_a R_b^2 R_c}{(R_a+R_b+R_c)^2}$$

$$R_2 R_3 = \frac{R_a^2 R_b R_c}{(R_a+R_b+R_c)^2}$$

$$R_1R_2+R_2R_3+R_1R_3 = \frac{R_a R_b R_c \cancel{(R_a+R_b+R_c)}}{(R_a+R_b+R_c)^{\cancel{2}}}$$

$$= \underset{\Delta}{R_c} R_3$$

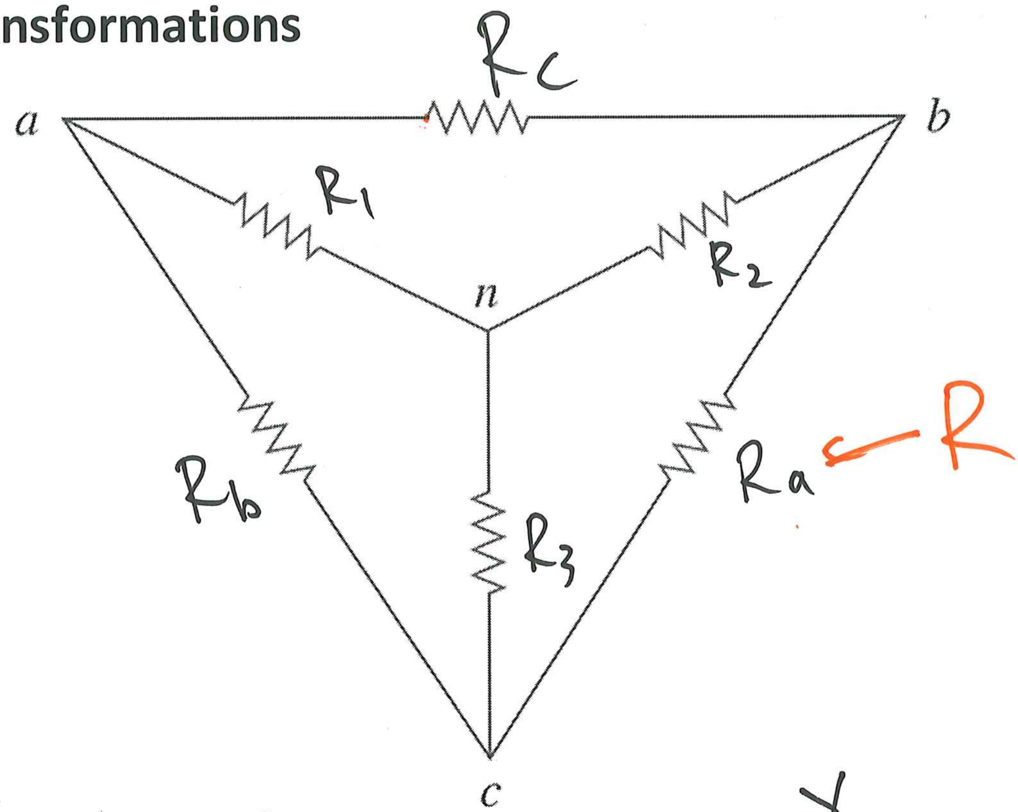
$$R_c = \frac{R_1R_2+R_2R_3+R_1R_3}{R_3}$$

$$R_a = \frac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_1}$$

$Y \to \Delta$

$$R_b = \frac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_2}$$

Transformations



$$\Delta \rightarrow Y$$

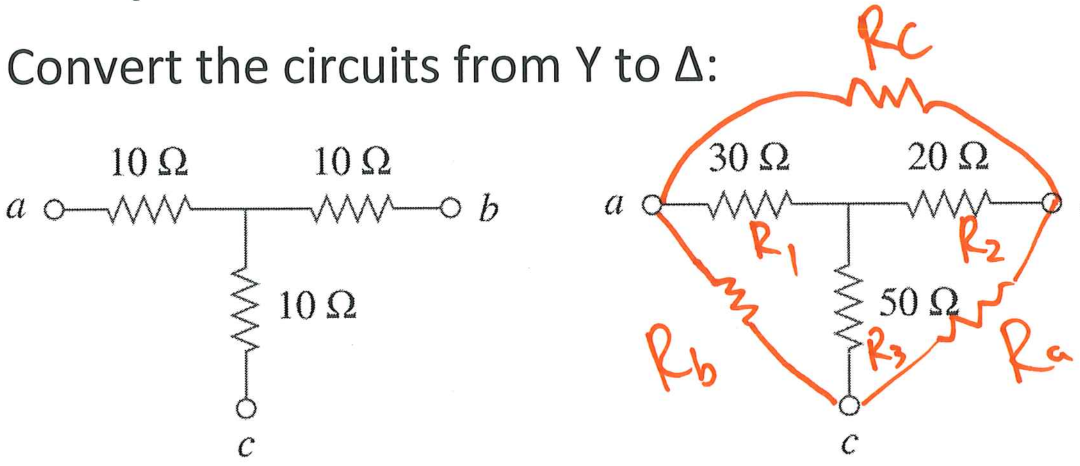
$$R_2 = \frac{R_a R_c}{R_a + R_b + R_c}$$

$$R_3 = \frac{R_a R_b}{R_a + R_b + R_c} = \frac{R^2}{3R} = \frac{R}{3}$$

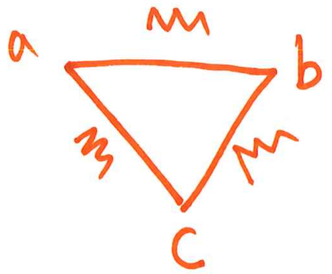
$$Y_R = \frac{\Delta_R}{3} \Rightarrow \Delta_R = 3 Y_R$$

Example #1:

Convert the circuits from Y to Δ:



(a) (b)



$\Delta_R = 3\Delta_y = 30\Omega.$

$$R_a = \frac{R_1R_2 + R_2R_3 + R_1R_3}{R_1}$$

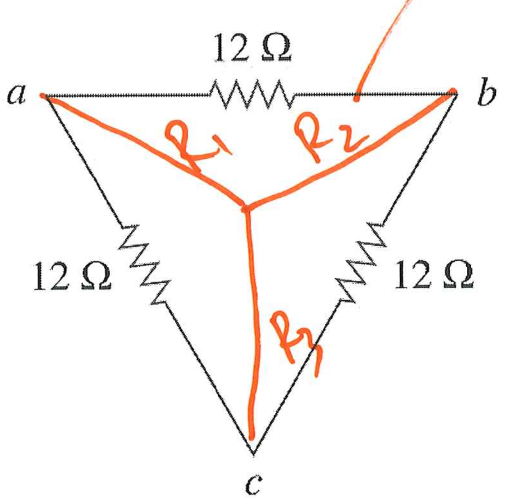
$$= 70 + \frac{50 \times 20}{30}$$

$$= 73.3\ \Omega$$

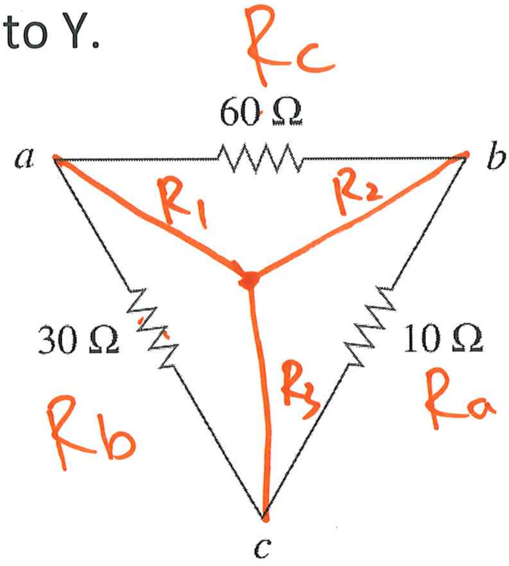
Example #2:

Convert the circuits from Δ to Y.

4Ω



(a)



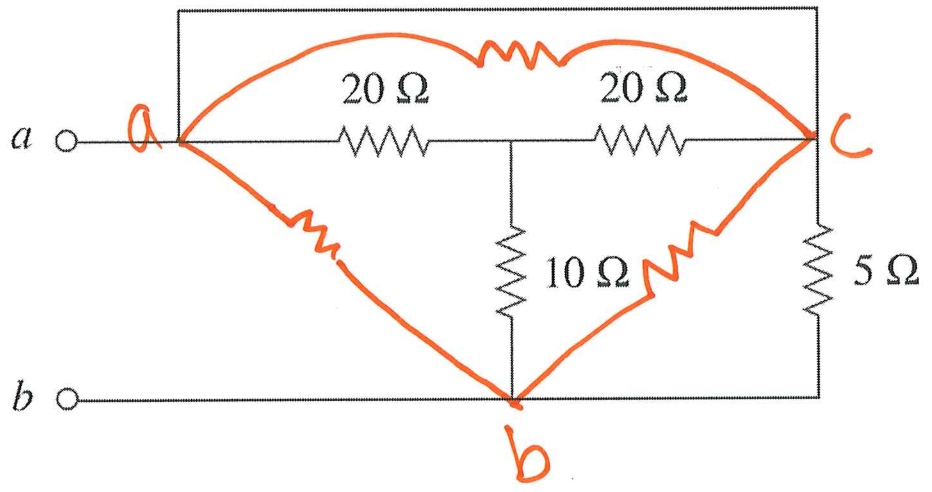
(b)

$$R_1 = \frac{30 \times 60}{30 + 60 + 10} = 18\Omega$$

R_2 ... R_3

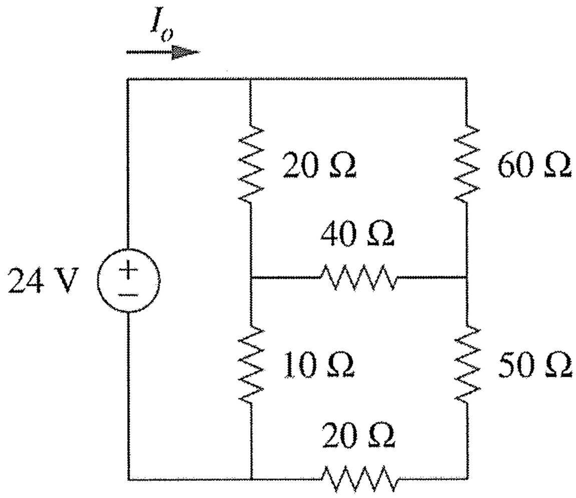
Example #3: homework

Determine R_{eq} between terminals *a* and *b*.



Example #4:

Calculate I_o in the circuit.



Example #3:

Determine i_1 to i_5.

homework